



Analysis of the Human Development Index (HDI)

A report on Data-Driven Exploration of Global and Regional Development Patterns

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INTRODUCTION

Short History of Human Development Index

Human Development Index (HDI) is a composite statistical measure that was created by the United Nations Development Programme (UNDP) to determine the level of development of countries. The HDI is a multidimensional measure of human well-being, which includes health, education, and standard of living, unlike traditional measures that only consider the economy, especially the Gross Domestic Product (GDP). Life expectancy at birth measures health, expected and mean years of schooling measures education and standard of living is measured by Gross National Income (GNI) per capita. The combination of the three dimensions makes the HDI a more holistic and people-focused approach to development, allowing comparative analyses of countries and time to evaluate progress and inequality.

Objectives of the Analysis

The main aim of this HDI analysis is to analyze and compare the level of human development of different countries and regions based on the available data of HDI. The analysis aims at establishing trends, patterns and difference in development over the years, therefore, pointing out areas that are improving and those that are still lagging. The study will also categorize countries based on specific levels of HDI, which are low, medium, high, and very high, as well as comparing the developmental patterns of South Asia and the Middle East..

Scope of the Report

The report uses the tools of exploratory data analysis (EDA) to compare and contrast the HDI values of various countries and regions visually.

Problem-Wise Analysis

Problem 1A - Single Year HDI Exploration:

Methods:

- Only the HDI data for the year 2022 was selected for detailed analysis.
- Developed classification of countries into development groups by particular HDI values.
- Refined and ranked countries by HDI scores.
- The dataset was thoroughly cleaned and imputed, ensuring there were no errors, duplicates, or missing values. All key descriptive statistics were calculated, such as the mean, median, and standard deviation.

Key Results:

- Classification of countries by their thresholds of HDI: 'Low' to 'Very High.'
- 69 Countries are classified into Very High HDI, while 35 countries got classified into Low HDI.

Visualizations and tables:

Figure 1 Statistical Summary

Mean: 0.72
Median: 0.74
Standard Deviation: 0.15

The data presents the 2022 Human Development Index (HDI) statistics, including the mean, median, and standard deviation. It also identifies the countries with the highest and lowest HDI, with Switzerland at the top and Somalia at the bottom.

Figure 2 Countries with (HDI > 0.8) sorted by GNI (2022)

[103]:

	country	year	hdi	gross_inc_percap	life_expectancy
3332	Liechtenstein	2022	0.942	146673.24150	84.656
4718	Qatar	2022	0.875	95944.37754	81.559
5213	Singapore	2022	0.949	88761.14559	84.133
2705	Ireland	2022	0.950	87467.51391	82.716
3398	Luxembourg	2022	0.927	78554.23640	82.591
6104	United Arab Emirates	2022	0.937	74103.71494	79.196
5609	Switzerland	2022	0.967	69432.78669	84.255
4322	Norway	2022	0.966	69189.76165	83.393
6170	United States	2022	0.927	65564.93798	78.203
1550	Denmark	2022	0.952	62018.95694	81.882

Figure 2 shows the top countries with (HDI>0.8) sorted by GNI per capita in the year 2022.

Qatar is observed to have the highest GNI per capita according to the table.

Figure 3 Table with HDI Categories:

	country	hdi	HDI_Category
32	Afghanistan	0.462	Low
65	Albania	0.789	High
98	Algeria	0.745	High
131	Andorra	0.884	Very High
164	Angola	0.591	Medium
197	Antigua and Barbuda	0.826	Very High
230	Argentina	0.849	Very High
263	Armenia	0.786	High
296	Australia	0.946	Very High
329	Austria	0.926	Very High

The tables illustrate the ranking of countries according to gross national income (GNI) per capita and their HDI classification. Countries like Qatar, Singapore, and Ireland fall into the high HDI group, while Yemen, Congo, and Somalia are placed in the low HDI group.

Interpretation and Discussion:

A clear pattern emerges showing a strong correlation between economic prosperity and human development, as countries with higher HDI often have higher GNI. Nevertheless, this is not always the case. Some countries may achieve a high HDI even with a relatively low GNI, since HDI also takes into account other dimensions such as education, health, and quality of life.

Problem 1B – HDI Trend Analysis (2020–2022)

Methods/ Approach:

- Selected HDI data covering the period from 2020 to 2022.
- Cleaned and structured the dataset to improve its accuracy and reliability.
- Categorized countries by region to make trends easier to compare and visualize.
- Applied various visualization techniques to examine HDI patterns across different regions and years.

Key Results:

- The COVID-19 pandemic significantly contributed to a decline in overall HDI during 2020–2022.
- Europe and Central Asia recorded the highest HDI values, while Sub-Saharan Africa had the lowest.

Visualizations and tables

Figure 4 Line graph for selected countries in (2020 - 2022)

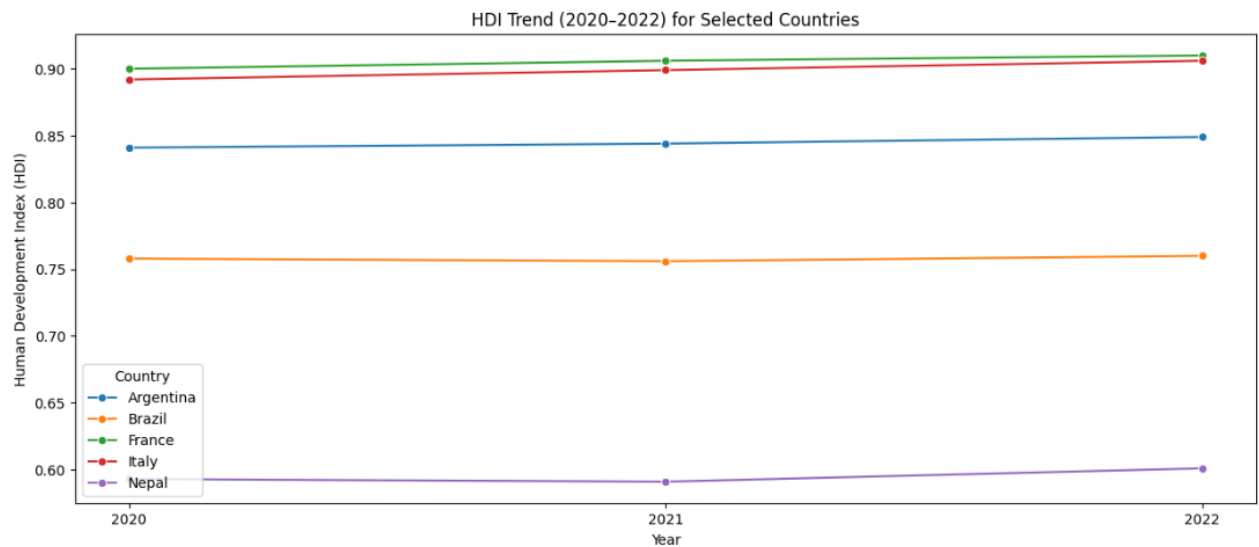
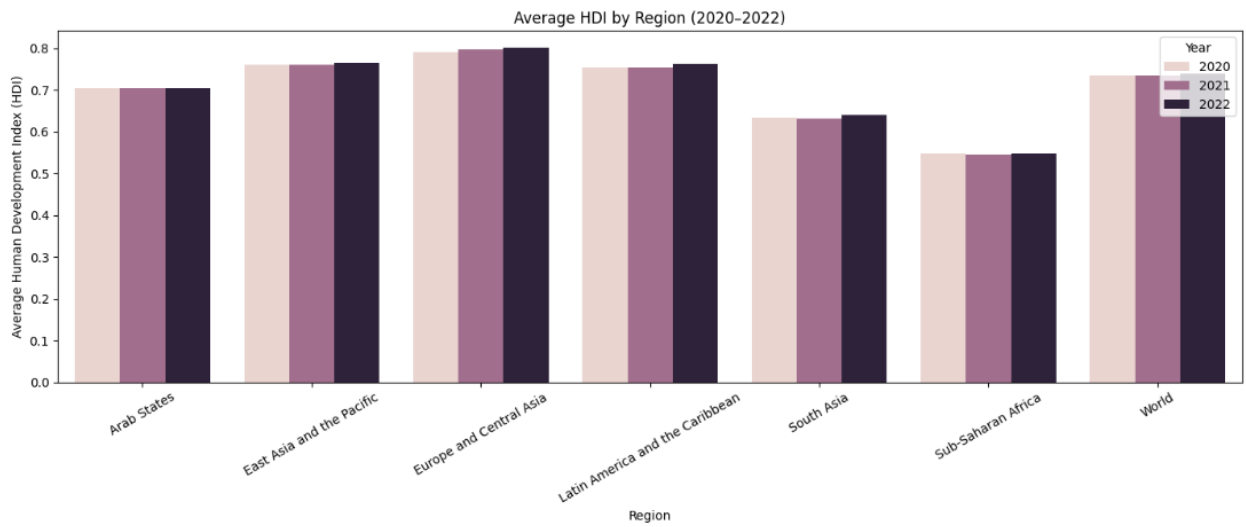


Figure: HDI trend across 2020-2022 shows progress or decline in human development.

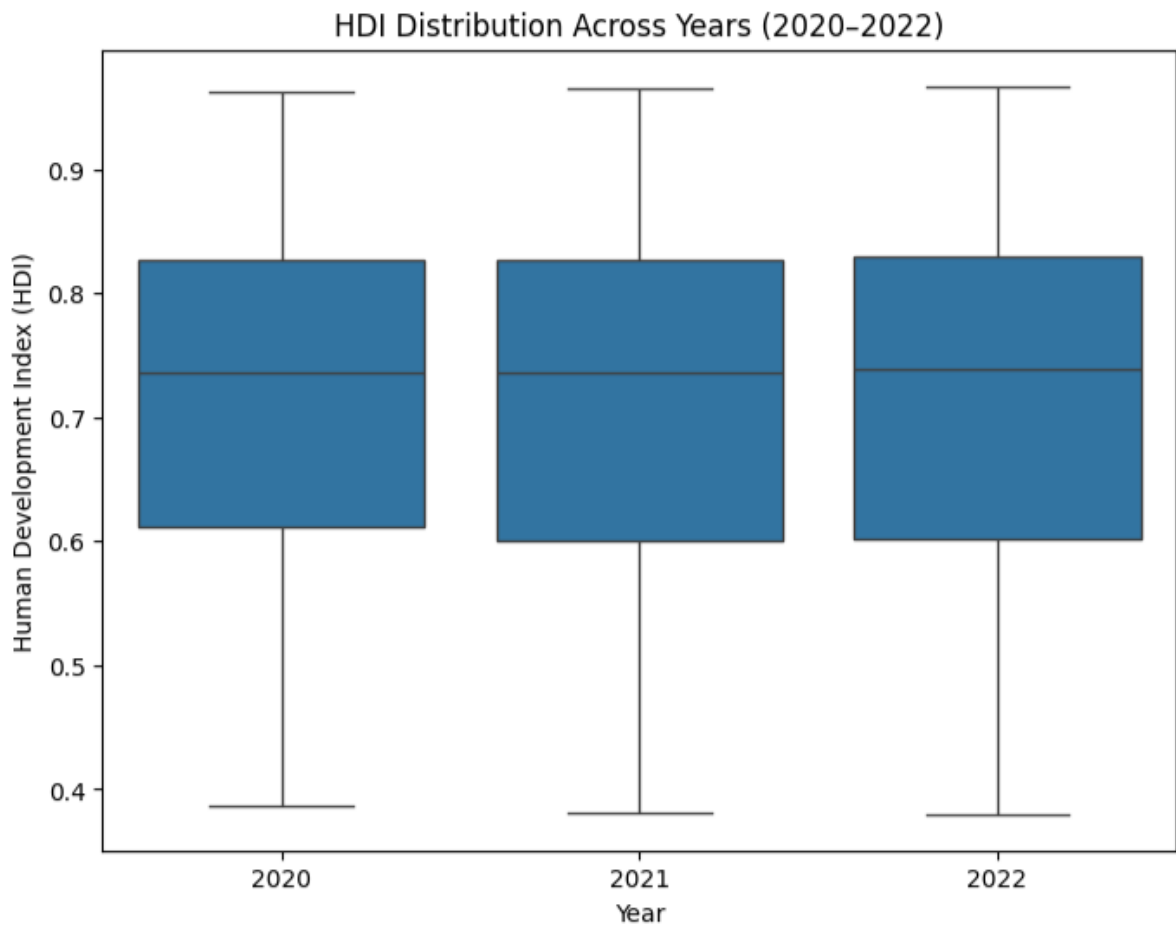
The line chart depicts the HDI progression of selected countries—Australia, China, India, Nepal, and Pakistan—over time. It clearly shows that Australia and China have higher HDI scores compared to India, Nepal, and Pakistan, indicating higher levels of human development.

Figure 5 Bar Chart Average HDI by Region



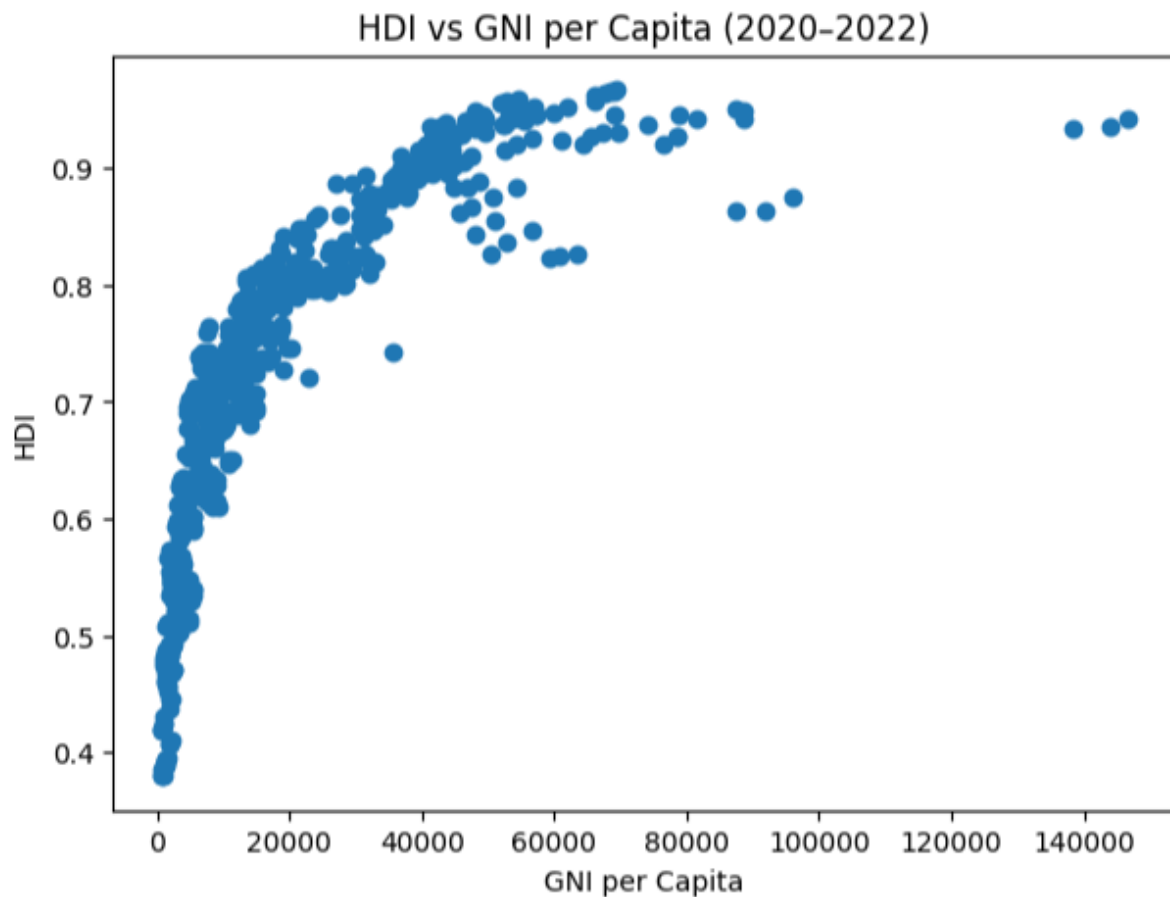
This bar chart represents the average HDI by region of 2020 to 2022. It demonstrates that Europe and Central Asia have the highest rate of HDI whereas Sub Saharan Africa has the lowest average HDI, in comparison to the other regions.

Figure 6 Box Plot: HDI Distribution for 2020, 2021, and 2022



The above boxplot presents the spread of HDI values for 2020, 2021, and 2022, highlighting the median, range, and variation for each country annually. The visualization suggests a gradual increase in HDI over the years, with 2022 showing a slightly higher overall distribution than 2020 and 2021.

Figure 7 Scatter Plot: HDI vs. GNI per Capita



The relationship between HDI and GNI per capita of 2020-2022 can be demonstrated by this scatter plot. It is clearly seen as having a positive correlation whereby the higher the income per capita of a country, the higher the HDI of the country tends to be on the average. The higher the GNI, the higher the human development outcomes of the countries.

Interpretation and discussion

The analysis of HDI trends from 2020 to 2022 shows noticeable variations in human development across different regions and countries. A strong positive correlation exists between Gross National Income (GNI) per capita and HDI, suggesting that higher income levels often support better development. However, outliers demonstrate that income alone does not determine development, as factors such as education, health, and living standards also play a crucial role in promoting balanced and sustainable growth.

Problem 2 – Advanced HDI Exploration

Methods/ Approach:

- Selected HDI data specifically for South Asian countries for detailed analysis.
- Cleaned and organized the dataset to ensure accurate and reliable results.
- Identified and examined outliers, as well as explored the relationships between key indicators.
- Performed a gap analysis between GNI and HDI to understand variations in development.
- Utilized various visualization techniques to compare trends and relationships across countries.

Key Results:

- The Maldives recorded the highest composite development score among South Asian nations.
- No countries were classified as outliers in either HDI or GNI per capita

Visualizations and tables:

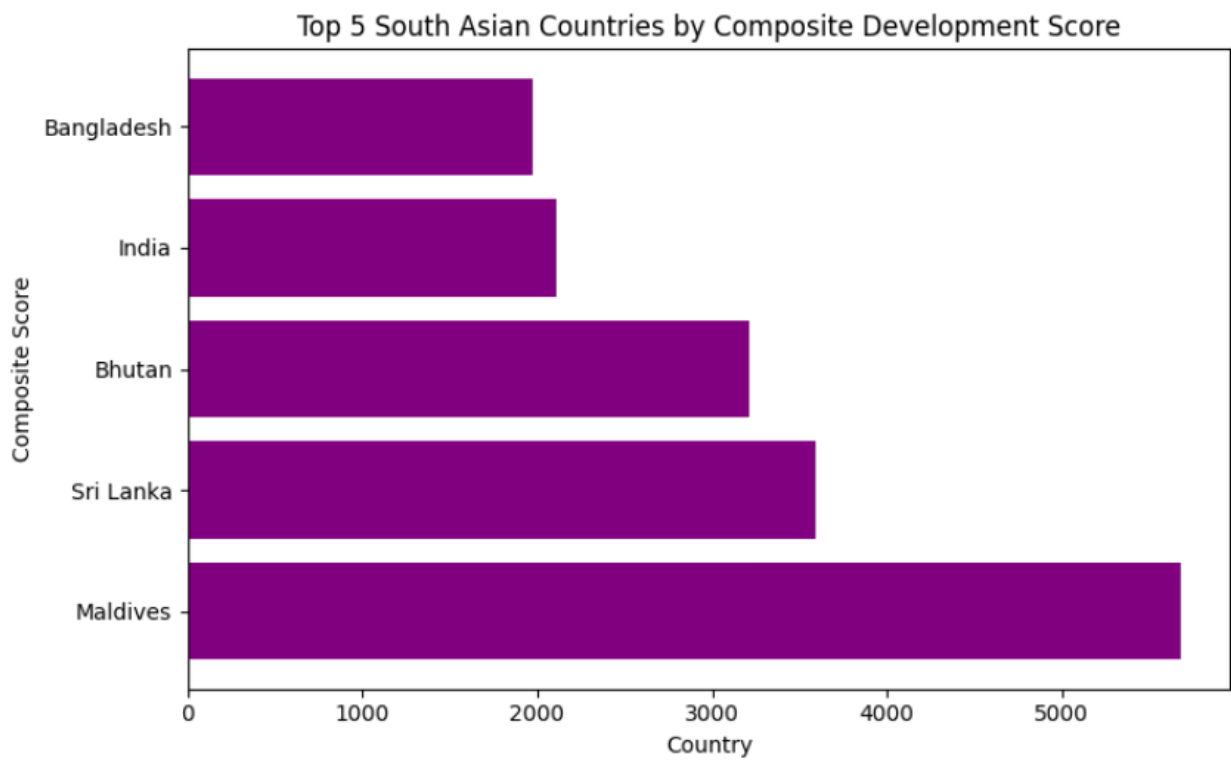
Figure 8 Table with composite score and HDI

Top 5 South Asian Countries by Composite Development Score:

	country	Composite_Score	Composite_Rank	hdi
164	Maldives	5678.289357	1.0	0.762
263	Sri Lanka	3592.832541	2.0	0.780
98	Bhutan	3209.130864	3.0	0.681
131	India	2105.481239	4.0	0.644
65	Bangladesh	1975.446053	5.0	0.670

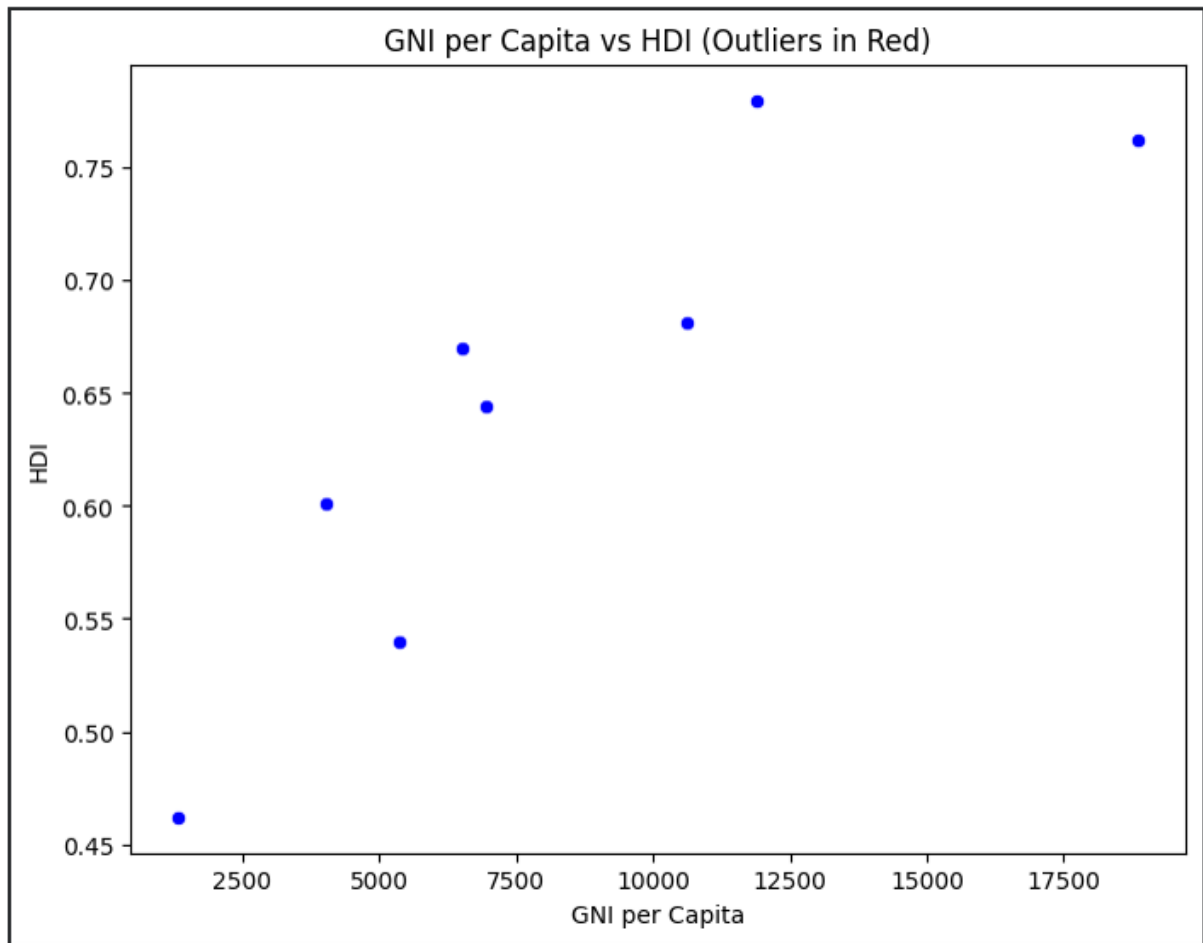
The table presents South Asian countries along with their composite development scores and HDI values. The Maldives ranks highest, reflecting a well-balanced development profile. Sri Lanka and Bhutan also show strong performance, whereas Afghanistan ranks lowest, indicating comparatively lower human development.

Figure 9 Bar graph of South Asian Countries by Composite Score



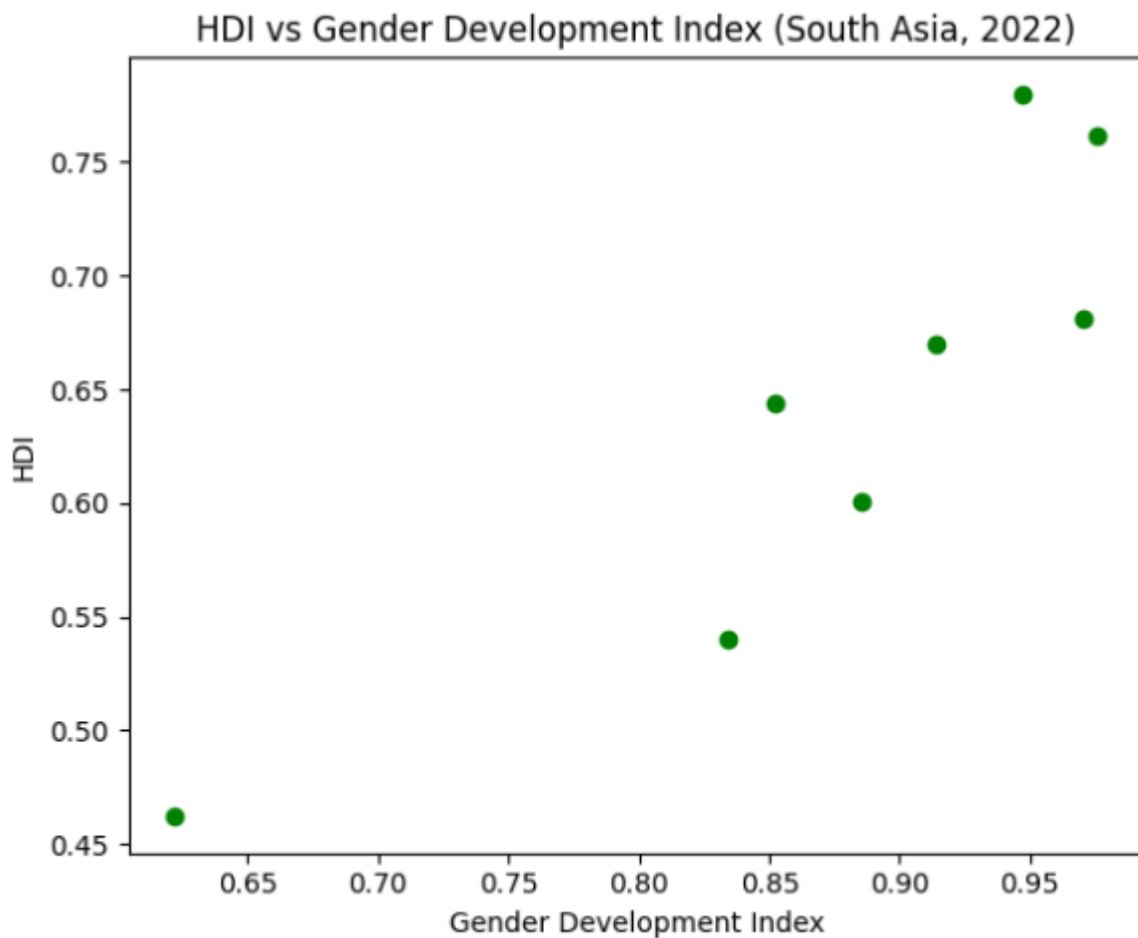
The bar chart reinforces this trend, showing the Maldives leading the region with a clear margin. Sri Lanka and Bhutan follow with relatively high development scores, while India and Bangladesh have lower scores, highlighting a noticeable gap between the top performers and other countries.

Figure 10 Scatter plot for GNI per capita vs HDI



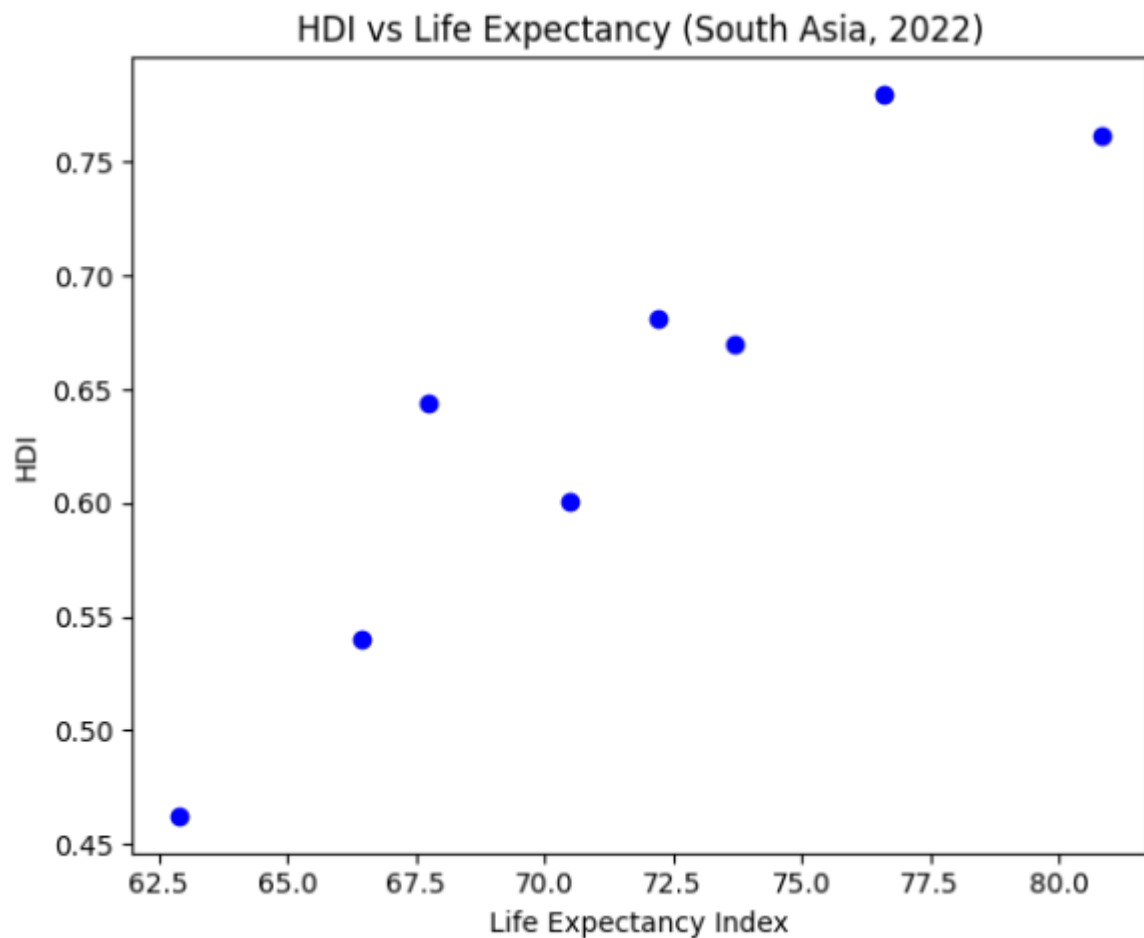
The scatter plot illustrates a positive relationship between GNI per capita and HDI across these nations. Generally, human development rises with income, exemplified by a country achieving a high HDI of nearly 0.80 with GNI per capita around 12,000. Conversely, the least developed country exhibits both low income and low HDI .

Figure 11 Scatter plot for Gender Development vs HDI



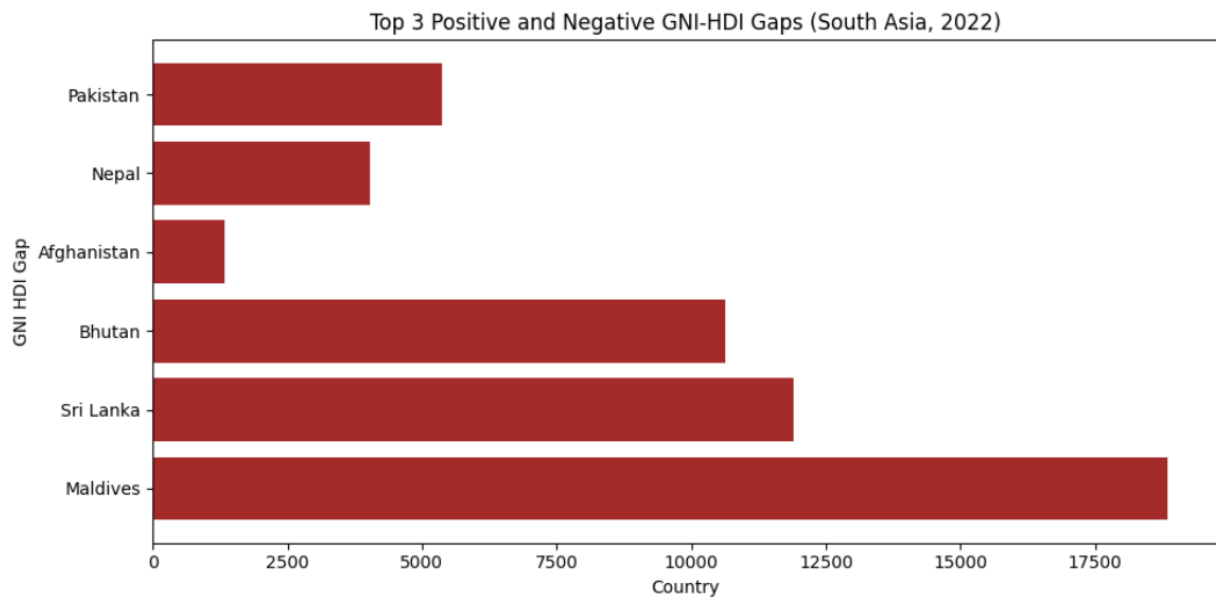
The scatter plot reveals a positive association between the Gender Development Index (GDI) and the Human Development Index (HDI) across countries. Improvements in gender equality—reflected in better health, education, and income for women—tend to be accompanied by higher overall human development. For example, one country shows a high HDI of nearly 0.78 with a GDI of about 0.95. On the other hand, the lowest-performing country exhibits both low gender development and low HDI, indicating that gender inequality can hinder overall development

Figure 12 Scatter plot for Life Expectancy vs HDI



Similarly, another scatter plot shows a positive correlation between life expectancy and HDI. Generally, countries with longer life expectancy have higher HDI values, suggesting that better health and longer lifespans contribute to human development. For instance, a country with an HDI around 0.78 and life expectancy of roughly 77 years demonstrates strong health standards, while the least developed country, with life expectancy near 63 years, has a low HDI, reflecting the link between shorter lifespans and lower development outcomes.

Figure 13 Bar chart showing Positive and Negative GNI-HDI



The bar chart compares the three countries with the largest positive and negative gaps between GNI and HDI, highlighting differences between income levels and human development outcomes. Countries such as the Maldives, Sri Lanka, and Bhutan show significant positive GNI-HDI gaps, indicating that increases in national income are accompanied by improvements in human development. In contrast, Pakistan, Nepal, and Afghanistan display smaller or negative gaps, suggesting that higher income does not necessarily translate into better human development. Overall, the chart emphasizes that per capita income alone does not guarantee human development, and countries vary in their ability to convert economic gains into broader prosperity.

Interpretation and discussion

The review of South Asian nations provides several important observations regarding economic and human development. No notable outliers were found in HDI or GNI per capita, suggesting that the trends are consistent. The strong positive relationships among HDI, life expectancy, and gender development emphasize the role of these factors in overall development. The Maldives consistently ranks highest in the composite development index, reflecting balanced growth in health, gender equality, and economic progress.

Problem 3 – Comparative Regional Analysis: South Asia vs Middle East

Methods/ Approach:

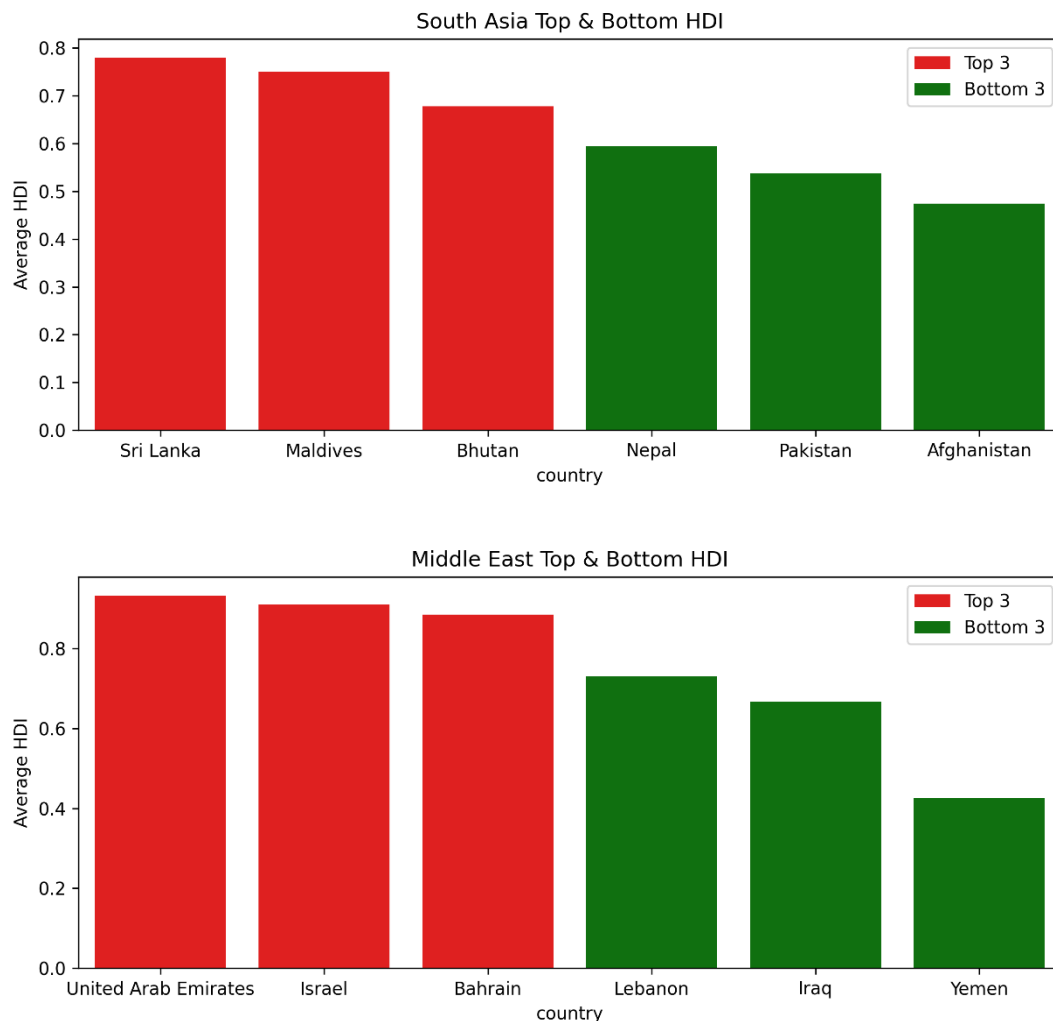
- Focused the HDI analysis on Middle Eastern and South Asian countries for the period 2020–2022.
- Cleaned and structured the data to ensure accurate and reliable analysis.
- Calculated descriptive statistics, including the mean and standard deviation.
- Performed outlier detection to identify extreme values.
- Compared HDI differences across countries and regions.
- Used multiple visualization techniques to analyze trends and relationships..

Key results

- Overall, Middle Eastern countries recorded higher HDI scores compared to South Asian nations between 2020 and 2022.
- The largest gap in GNI per capita was observed between South Asia and the Middle East.
- Positive trends were found across the analyzed relationships, indicating that higher gender development and longer life expectancy are associated with increased HDI levels.

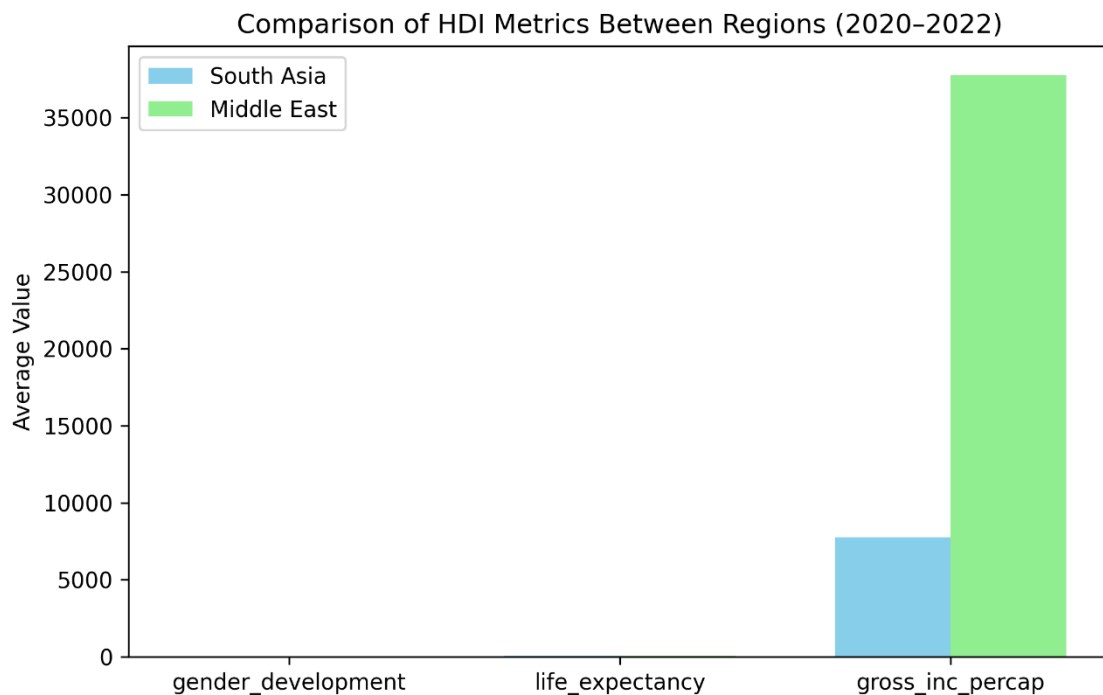
Visualizations and tables

Figure 14 Bar graph with Top & Bottom HDI Performers (South Asia vs Middle East)



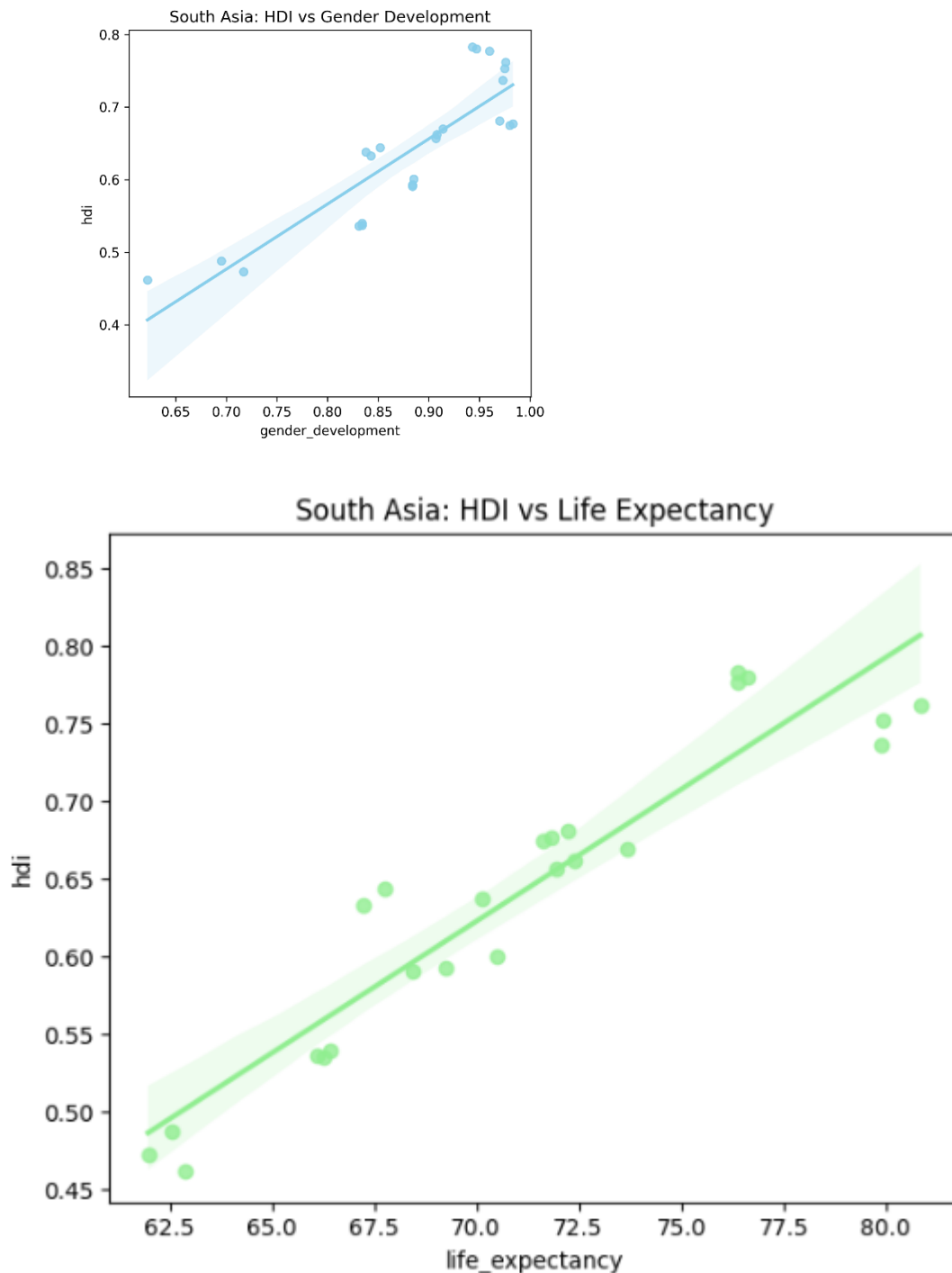
The bar graph comparing HDI scores between South Asia and the Middle East clearly illustrates regional differences. Middle Eastern countries like the United Arab Emirates, Israel, and Bahrain achieve higher HDI values, reflecting stronger economies, better healthcare, and higher living standards. South Asian countries show greater variation: Sri Lanka and the Maldives perform relatively well, while Afghanistan and Pakistan rank at the lower end. Overall, the graph indicates that higher income levels, education, and healthcare availability contribute to better human development in the Middle East, although some exceptions exist..

Figure 15 Bar graph for Metric Comparison Between Regions



Another bar graph comparing key development indicators—gender development, life expectancy, and GNI per capita—highlights similar regional disparities. South Asian nations consistently score lower than Middle Eastern countries across all three measures. The greatest difference is in GNI per capita, emphasizing the stronger economic conditions and higher standards of living in the Middle East. Higher scores in gender development and life expectancy also reflect better access to education, healthcare, and social services. In summary, the graphs suggest that economic prosperity is closely tied to improved human and social development, while South Asia continues to face structural and economic challenges, despite gradual progress.

Figure 16 : Scatter plot showing South Asia: Gender Development vs HDI & Life Expectancy vs HDI

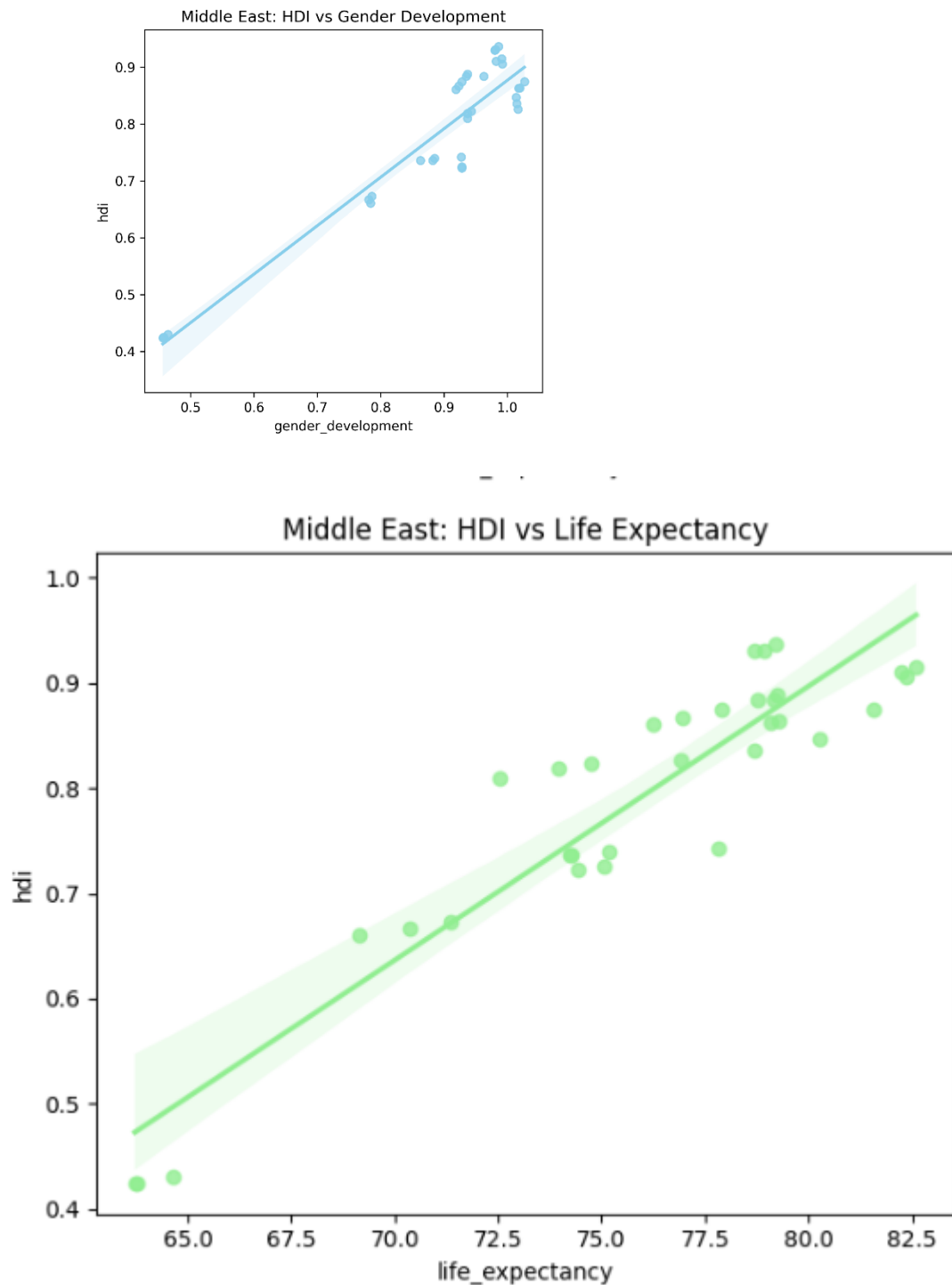


The scatter plots for South Asia show a strong positive correlation between HDI and two key human development factors: gender development and life expectancy. The first plot indicates that countries with higher gender development—where women have greater access to education and healthcare—tend to have higher HDI scores. This demonstrates that gender equality and female participation in education and health are closely linked to overall human development. While a few countries deviate slightly from the trend, the general upward pattern

confirms this relationship.

Similarly, the second plot shows that higher life expectancy is associated with higher HDI. Countries with longer average lifespans consistently record better HDI scores, highlighting the impact of healthcare, nutrition, and living standards on development outcomes. Together, these plots suggest that in South Asia, improvements in social factors like gender equality and health are closely tied to higher human development, although national disparities cause minor deviations from the trend.

Figure 17 Scatter plot showing Middle East: Gender Development vs HDI & Life Expectancy vs HDI



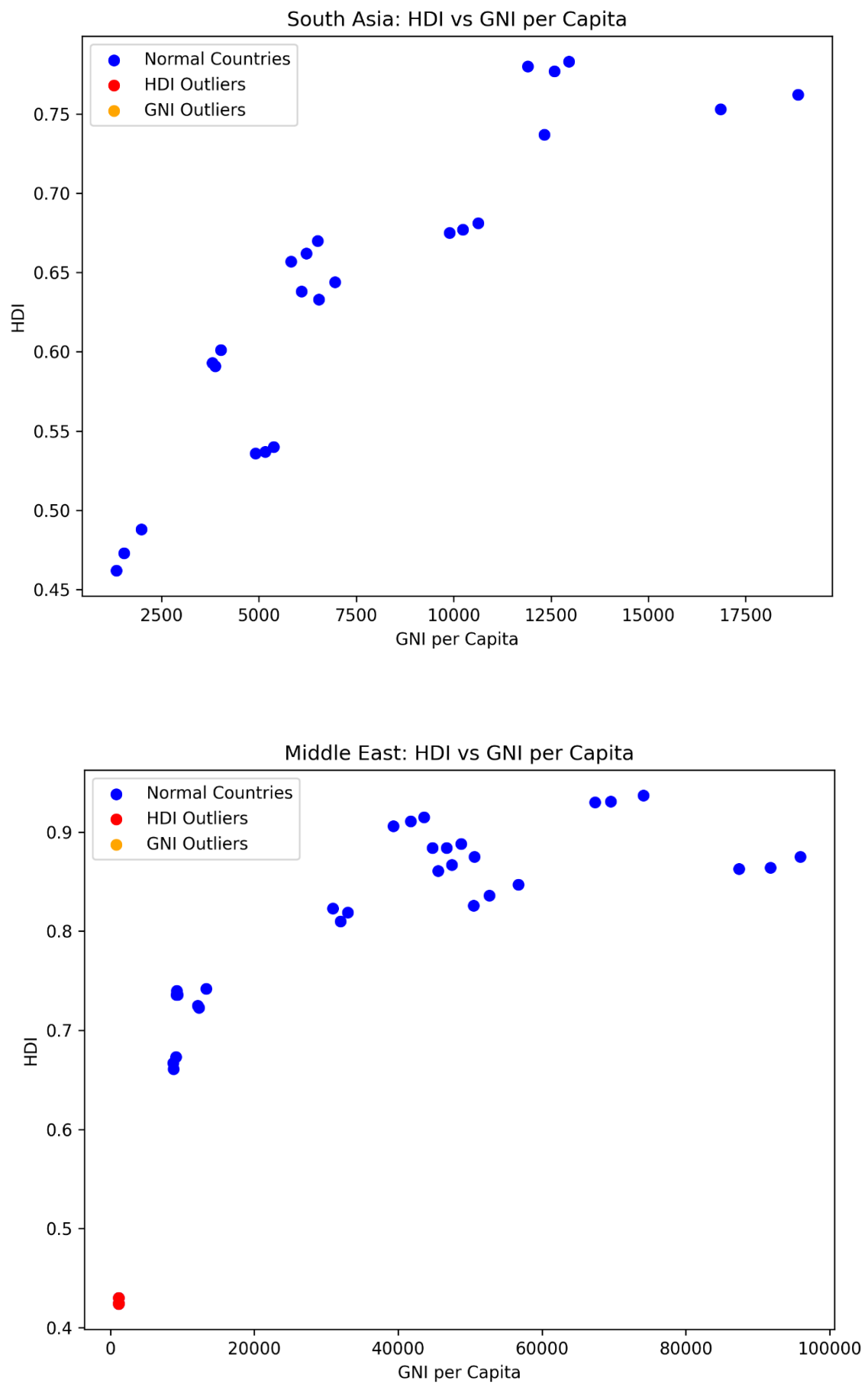
Gender Development vs HDI

In the Middle East, a similar pattern emerges. The scatter plot illustrates a positive relationship between gender development and HDI, showing that countries with greater gender equality—through enhanced opportunities for women in education, healthcare, and economic participation—achieve better human development outcomes.

Life Expectancy vs HDI

Additionally, life expectancy also positively correlates with HDI in the region. Countries with higher average life spans tend to have higher HDI scores, underscoring the importance of healthcare, nutrition, and living conditions. Overall, improved health outcomes play a key role in enhancing human development levels across the Middle East.

Figure 18 Scatter plot showing South Asia and Middle East Outliers:



The outlier scatter plots for South Asia and the Middle East highlight countries that deviate from the general trend between gross national income (GNI) per capita and HDI. In South Asia, some countries achieve relatively high HDI despite low income, suggesting that effective social policies, education, or healthcare may help offset lower economic resources. Conversely, certain wealthier countries show lower HDI, indicating uneven or unbalanced development.

In the Middle East, the pattern contrasts sharply: while there are several high-income countries with very high HDI, the region also contains an extreme outlier with both low income and low HDI. This reflects significant income inequality and demonstrates that although higher income can support HDI, governance and social investment play a critical role in translating economic resources into human development.

Interpretation and discussion

The analysis of South Asian and Middle Eastern countries from 2020 to 2022 shows that Middle Eastern nations generally perform better in HDI, mainly due to higher gender equality and stronger economic conditions. Overall, HDI is closely linked with life expectancy, gender development, and economic prosperity. Although some outliers and minor imbalances exist, they do not significantly affect the overall positive trend.

CONCLUSION

Summary of Key Findings

There exists a strong positive correlation indicating that as the HDI index increases, the Gross National Income (GNI) per capita also tends to rise; however, some notable exceptions are present. The effects of the pandemic have negatively influenced various fundamental aspects of the HDI, hindering overall developmental progress. On average, countries in the Middle East exhibit better HDI performance compared to those in South Asia, which is due to improved gender development indicators and higher income levels, even though a few outliers exist..

Examples of the HDI Trends and Disparities

Significant disparities in Human Development Index (HDI) levels exist across regions and countries. The HDI is generally positively correlated with key development indicators such as gender equality, per capita income, health outcomes, and educational attainment. Some countries exhibit over-performance or under-performance due to outlier effects, inefficient resource utilization, or governance challenges, indicating that development potential is not fully realized. While the Middle East typically demonstrates higher average HDI scores than South Asia, it also displays greater inter-country variation, reflecting unequal development and management inefficiencies.

Limitations of the Analysis

The present analysis utilizes datasets spanning 1990 to 2022, which may not accurately reflect current conditions. While focusing on the most recent three years provides relevant insights, limiting the assessment to short-term data restricts the ability to identify broader historical trends.

Implications and Recommendations

Differences in HDI performance are often linked to weak policy implementation and inadequate governance in certain countries. Strengthening institutional accountability, adopting effective policies, and using development data more strategically can help identify areas that require improvement. These measures can support more balanced and sustainable progress in Human Development Index outcomes in the coming years.

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Assignment PDF / Guidelines

Assignment - 1 – Statistical Interpretation and Exploratory Data Analysis – 2025

APPENDIX

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